SHAHADAT HUSSAIN, PhD

Research Associate · Division of Engineering (CTP Lab) · New York University Abu Dhabi
+971 50 6280812 | shahadat_ind@yahoo.com |
shahadathussain.com | linkedin.com/in/shahadathussain
ORCID: 0000-0002-4355-2169 | Scopus: 56380929800 | ResearcherID: I-3091-2017 | ISNI: 0000 0005 3020 7229

PROFILE SUMMARY

Research Associate at New York University Abu Dhabi specializing in metamaterials, additive manufacturing, and shape memory alloys (NiTi and Cu-based). Expertise in 3D printing of triply periodic minimal surface (TPMS) lattices, microstructural analysis, mechanical characterization, and data-driven materials research. Proven track record of high-impact publications in reputed journals and international conferences, with strong skills in scientific computing (MATLAB and Python), data science, and advanced manufacturing techniques. Passionate about developing next-generation smart materials and architected structures for mechanical and structural applications. Active peer reviewer for 15+ Elsevier and Springer journals.

RESEARCH METRICS

500+ Citations	**h-index: 10**	**i10-index: 10**	**15+ Publications**

Source: Google Scholar (May 2026).

RESEARCH INTERESTS

Metamaterials & TPMS lattices · Fused Filament Fabrication (FFF) / Stereolithography (SLA) / Selective Laser Sintering (SLS) / Laser Powder Bed Fusion (LPBF) / Additive Manufacturing · NiTi & Cu-based shape memory alloys · Fatigue testing · Microstructural characterization · Thermal analysis · High-temperature casting · Hot rolling · Heat treatment · Scientific computing & data science

TEACHING INTERESTS

Engineering Thermodynamics · Mechanics of Materials · Materials Science · Manufacturing Processes

EDUCATION

Doctor of Philosophy (PhD) — Materials Science and Technology | **Academy of Scientific and Innovative Research (AcSIR) | CSIR-AMPRI**

August 2014 – July 2019 · Bhopal, Madhya Pradesh, India

- Doctoral research on Cu-Al-Ni shape memory alloys; studied impact of grain refiners, alloying additions, and processing parameters on shape memory properties for high-temperature applications.
- Awarded Highest CGPA in PhD Coursework (2016).
- Completed 14 subjects during coursework (2014–2016): Materials Science & Engineering, Phase Transformation, Functional & Smart Materials, Nano Science & Engineering, Composite Science, Lightweight Materials, Powder Metallurgy, Computer Simulation & Design, and more — achieving distinction.
- Trained lab operators in casting shop on vacuum induction melting and die casting; created standard operating procedure manual for furnace operation.
- Conducted CSIR-800 entrepreneurship study in a village involving surveying, questionnaire design, data collection, analysis, and report generation.
- Published 2 research articles; delivered 5 conference presentations.
- Collaborated with Prof. V. Sampath (IIT-Madras) on induction melting and Cu-based SMA synthesis.
- Thesis available at: http://hdl.handle.net/10603/384933

Bachelor of Engineering (BEng) — Mechanical Engineering | **University Institute of Technology – RGPV**

August 2008 – June 2012 · Bhopal, Madhya Pradesh, India

- Graduated with First Division aggregate.
- Two-week industrial training at SAIL-Bokaro: practical knowledge in hot rolling, cold rolling, continuous casting, machining, foundry, steel melting, blast furnace, coke oven, thermal power plant and.

WORK EXPERIENCE

Research Associate | Division of Engineering (CTP Lab), New York University Abu Dhabi

September 2024 – Present · Abu Dhabi, UAE

- Hybrid manufacturing including additive manufacturing via stereolithography, fused deposition modelling, polyjet printing, and investment casting.
- Conducting compression and bending testing; characterization using SEM, XRD, EDS, and metallography.
- Investigating bending analysis of single-phase and interpenetrating phase composite metamaterial (Primitive and Gyroid) cubes and beams.
- Advanced data analysis and visualization using OriginLab, MATLAB and toolboxes, Python (pandas, matplotlib, numpy, scipy, scikit-learn, seaborn).

Postdoctoral Researcher | Department of Mechanical Engineering, Khalifa University

February 2020 – June 2024 · Abu Dhabi, UAE

- Examined and characterized properties of additively manufactured NiTi TPMS structures (Schwartz Primitive and Schoen Gyroid lattices) — porous, cellular, and architected materials.
- Performed tensile and fatigue tests on thin NiTi wires with different diameters, chemical compositions, and surface treatments (etched, mechanically polished, etched-polished, light oxide coating) using Instron 8872 fatigue testing machine.
- Applied advanced analytical techniques: optical microscopy, SEM, XRD, EDS, DSC, metallography, heat treatment, EBSD (Electron Backscatter Diffraction), AFM (Atomic Force Microscopy).
- Collaborated with Prof. Greg Haidemenopoulos (University of Thessaly, Volos, Greece) on microstructural characterization of LPBF-fabricated NiTi architected alloys.
- Completed 10-week Teaching Certificate Course, Center for Teaching and Learning (CTL), Khalifa University (2023).
- Completed Responsible Conduct of Research (RCR) — CITI Program, Florida, USA (2020), achieving 100% grade.
- Completed Lab Safety & EHS certification (Khalifa University, 2021): lab safety, compressed gas, radiation, chemical safety, cryogenic substances, HF acid handling, electrical safety, fire safety, waste management.

Project Fellow | CSIR – Advanced Materials and Processes Research Institute (AMPRI)

May 2013 – August 2014 · Bhopal, Madhya Pradesh, India

- Extensive research experience in alloy synthesis, metallography, heat treatment, and material characterization using spectroscopy, microscopy, diffractometry, calorimetry, hardness, tensile testing, and hot rolling.
- Published 5 research articles; presented research via oral and poster presentations; collaborated with IIT-Madras.

Lecturer — Mechanical Engineering | Oriental Group of Institutions

September 2012 – May 2013 · Bhopal, Madhya Pradesh, India

- Taught mechanical engineering subjects: materials science, machine component design, mechanics of materials.
- Participated in ISTE workshop on Engineering Thermodynamics by IIT Bombay; positive student feedback; completed syllabus on schedule.

RESEARCH PROJECTS

Additive Manufacturing of NiTi Shape Memory Alloys: Constitutive Modeling and Fatigue Failure Criteria | Khalifa University

February 2020 – June 2024 · Abu Dhabi, UAE

- 3D printing of NiTi samples and TPMS structures using EOS M400 LPBF system.
- Extensive characterizations: SEM, TEM, XRD, AFM, DSC, and fatigue testing.
- Novel microstructural and phase transformation findings published in high-impact journals.
- Tools: Additive Manufacturing, LaTeX/Overleaf, Python, MATLAB.

Design and Development of Thermo-Responsive and Magnetic Shape Memory Materials and Devices | **CSIR-AMPRI**

June 2013 – August 2014 · Bhopal, Madhya Pradesh, India

- Developed copper-based shape memory materials with superior mechanical properties, high transition temperatures, and cost-effective production of shape memory wires and strips.
- Outcomes: 5 journal articles, national conference presentations, collaboration with IIT-Madras.
- Tools: Hot Rolling, High Temperature Casting, Heat Treatment, Spectroscopy, Microscopy, Mechanical Testing, Origin, ImageJ.

JOURNAL PUBLICATIONS

- Hussain S, Mehraj H, Karathanasopoulos N, Moustafa MA (2026). Do chopped fibers matter? Process-Structure-Property Relationships and Thermal Annealing Effects in Fiber-Reinforced TPMS Lattices. Progress in Additive Manufacturing.
- Hussain S, Alagha AN, Zaki W (2024). Phase Transformation Behavior of NiTi Triply Periodic Minimal Surface Lattices Fabricated by Laser Powder Bed Fusion. Journal of Materials Engineering and Performance. https://doi.org/10.1007/s11665-024-09162-7
- Hussain S, Alagha AN, Haidemenopoulos GN, Zaki W (2023). Microstructural and Surface Analysis of NiTi TPMS Lattice Sections Fabricated by Laser Powder Bed Fusion. Journal of Manufacturing Processes, 102:375–386. https://doi.org/10.1016/j.jmapro.2023.07.055
- Hussain S, Alagha AN, Zaki W (2022). Imperfections Formation in Thin Layers of NiTi Triply Periodic Minimal Surface Lattices Fabricated Using Laser Powder Bed Fusion. Materials, 15(22):7950. https://doi.org/10.3390/ma15227950
- Alagha AN, Hussain S, Zaki W (2021). Additive Manufacturing of Shape Memory Alloys: A Review with Emphasis on Powder Bed Systems. Materials & Design, 204:109654. https://doi.org/10.1016/j.matdes.2021.109654
- Hussain S, Pandey A, Dasgupta R (2021). Nano-CeO2 Doped Cu-Al-Ni SMAs with Enhanced Mechanical as well as Shape Recovery Characteristics. Metals and Materials International, 27:1478–1482. https://doi.org/10.1007/s12540-019-00570-2
- Pandey A, Hussain S, Nair P, Dasgupta R (2020). Influence of Niobium and Silver on Mechanical Properties and Shape Memory Behavior of Cu-12Al-4Mn Alloys. Journal of Alloys and Compounds. https://doi.org/10.1016/j.jallcom.2020.155266
- Dasgupta R, Pandey A, Hussain S, Jain AK, Ansari A, Sampath V (2019). Effect of Microstructure on Roll-Ability and Shape Memory Effect in Cu-Based Shape Memory Alloys. Applied Innovative Research, 1:29–37.
- Hussain S, Pandey A, Dasgupta R (2019). Designed Polycrystalline Ultra-High Ductile Boron Doped Cu-Al-Ni Based Shape Memory Alloy. Materials Letters, 240:157–160. https://doi.org/10.1016/j.matlet.2018.12
- Pandey A, Jain AK, Hussain S, Sampath V, Dasgupta R (2016). Effect of Nano CeO2 Addition on the Microstructure and Properties of a Cu-Al-Ni Shape Memory Alloy. Metallurgical and Materials Transactions B, 47:2205–2210. https://doi.org/10.1007/s11663-016-0691-0
- Jain AK, Hussain S, Kumar P, Pandey A, Dasgupta R (2015). Effect of Varying Al/Mn Ratio on Phase Transformation in Cu-Al-Mn Shape Memory Alloys. Transactions of Indian Institute of Metals, 69:1289–1295. https://doi.org/10.1007/s12666-015-0689-3
- Hussain S, Kumar P, Jain AK, Pandey A, Dasgupta R (2015). Effects of Different Quaternary Additions in the Properties of a Cu-Al-Mn Shape Memory Alloy. Materials Performance and Characterization, 4:225–235. https://doi.org/10.1520/MPC20150017
- Kumar P, Jain AK, Hussain S, Pandey A, Dasgupta R (2015). Changes in the Properties of Cu-Al-Mn Shape Memory Alloy Due to Quaternary Addition of Different Elements. Revista Materia, 20:284–292. https://doi.org/10.1590/S1517-707620150001.0028
- Dasgupta R, Jain AK, Kumar P, Hussain S, Pandey A (2015). Role of Alloying Additions on the Properties of Cu-Al-Mn Shape Memory Alloys. Journal of Alloys and Compounds, 620:60–66. https://doi.org/10.1016/j.jallcom.2014.09.047
- Dasgupta R, Jain AK, Kumar P, Hussain S, Pandey A (2014). Effect of Alloying Constituents on the Martensitic Phase Formation in Some Cu-Based SMAs. Journal of Materials Research and Technology, 3:264–273. https://doi.org/10.1016/j.jmrt.2014.06.004

CONFERENCE PUBLICATIONS

- Hussain S, Alagha AN, Zaki W (2022). Inhomogeneous Microstructure due to Non-Uniform Solidification Rate in NiTi Triply Periodic Minimal Surface (TPMS) Structures Fabricated via Laser Powder Bed Fusion. ASME IMECE-2022, Columbus, Ohio, USA. https://doi.org/10.1115/IMECE2022-95320
- Hussain S, Jain AK, Ansari MA, Pandey A, Dasgupta R (2017). Study of Effect of Fe, Cr and Ti on the Martensite Phase Formation in Cu-12.5 wt%Al-5 wt%Mn SMA. Advanced Materials Proceedings, 2(1):22–25. https://doi.org/10.5185/amp.2017/106

BOOK CHAPTER

- Dasgupta R, Jain AK, Hussain S, Pandey A, Sampath V (2018). Effect of Alloying Additions on the Properties Affecting Shape Memory Properties of Cu-12.5Al-5Mn Alloy. In: Frontiers in Materials Processing, Applications, Research and Technology. Springer, Singapore. https://doi.org/10.1007/978-981-10-4819-7_33

CONFERENCE PRESENTATIONS

Oral Presentations:

- Best Practices in Teaching and Learning Conference, Khalifa University, Abu Dhabi (May 2023). Organized by Khalifa University, American University of Sharjah, and Amity University Dubai.
- "Inhomogeneous Microstructure due to Non-Uniform Solidification Rate in NiTi TPMS Structures Fabricated via LPBF" — ASME IMECE-2022, Columbus, Ohio, USA (November 2022).
- "Improvement in Shape Memory Properties of CuAlNiZn SMAs Due to Addition of Grain Refiners" — National Seminar on Advances in Smart & Functional Materials, CSIR-AMPRI, Bhopal, India (January 2017).
- "Study of Effect of Fe, Cr and Ti on the Martensite Phase Formation in Cu-12.5 wt%Al-5 wt%Mn SMA" — International Conference on Materials Science and Technology, University of Delhi, India (March 2016).

Poster Presentations:

- "Microstructural Changes and its Effect on Mechanical as well as Shape Memory Properties in a CuAlNi Alloy on Addition of Dispersoid" — RIAM-2018, CSIR-AMPRI, Bhopal (September 2018).
- "Changes in Properties Due to Alloying Addition of Mn and Cr to Cu-Al-Ni-Zn SMAs" — India International Science Festival (IISF-2016), CSIR-NPL, New Delhi (December 2016).
- "Effect of Addition of Alloying Element Chromium to Cu-Al-Ni-Zn Shape Memory Alloys" — 70th IIM NMD-ATM, IIT Kanpur (November 2016).
- "Role of Grain Refiners for Improved Shape Memory Properties of Cu-12.5 wt%Al-5 wt%Mn SMA" — National Seminar on Recent Innovations in Advanced Materials, CSIR-AMPRI, Bhopal (December 2015).

AWARDS, FELLOWSHIPS, GRANTS & HONORS

2016	Highest CGPA — PhD Coursework	*India*
2016	Senior Research Fellowship (SRF), CSIR-GATE	*India*
2014	Junior Research Fellowship (JRF), CSIR-GATE	*India*
2013	Qualified Graduate Aptitude Test in Engineering (GATE) — (2013, 2014, 2015)	*India*

PEER REVIEW ACTIVITY

Active reviewer for 15+ journals. Selected list:

- Additive Manufacturing; Additive Manufacturing Letters
- Materials & Design; Materials Characterization; Materials Science and Engineering A
- Journal of Manufacturing Processes; Journal of Materials Science and Technology
- Journal of the Mechanical Behavior of Biomedical Materials; Journal of Materials Research and Technology
- International Journal of Mechanical Sciences; Thin-Walled Structures
- Materials Letters; Materials Today Communications; Scripta Materialia
- Advances in Industrial and Manufacturing Engineering; Intermetallics

TECHNICAL EXPERTISE & INSTRUMENTATION

Additive Manufacturing / 3D Printing:

- FFF/FDM: Raise3D Pro3 Plus HS, Raise3D Pro2 Plus, Raise3D Forge1, Ultimaker, Stratasys F370
- PolyJet: Stratasys J750 / J850; SLA: Formlabs Form3+; FFF (high-speed): Bambu Lab Carbon X1
- Metal LPBF/SLM: EOS M400 (NiTi TPMS lattices)
- Slicing Software: Cura, GrabCAD Print, PreForm, IdeaMaker, Materialise Magics, Bambu Studio

Materials Characterization:

- SEM/EDS/FIB: JEOL 7610F, FEI Nova NanoSEM, FEI Quanta 3D FIB-SEM
- XRD: Bruker D2 Phasor, Rigaku MiniFlex II; Analysis: Xpert HighScore Plus
- DSC: Mettler Toledo DSC 1, Setaram Instruments
- EBSD (Electron Backscatter Diffraction), AFM (Atomic Force Microscopy)
- OES: Bruker Q4 Tasman; Optical Microscopy: Leica; Image analysis: Gatan Suite, ImageJ/Fiji

Mechanical Testing & Processing:

- Instron 5969 Static UTM (50 kN), Instron 8872 Fatigue (25 kN), MTS (100 kN)
- Controls Simplex 350, Controls Automax Compression Machine
- Vacuum induction melting, hot/cold rolling (asymmetric two-roll mill), heat treatment (Carbolite muffle furnaces), quenching
- Metallography: Struers/Buehler grinder-polisher; chemical etching including HF acid

CAD & Simulation:

- FreeCAD, SolidWorks, TinkerCAD

Scientific Computing & Data Science:

- Languages: Python, MATLAB (+ toolboxes), SQL (SQLite3), Shell scripting (Linux/Bash), LaTeX/Overleaf
- Data Science: Pandas, NumPy, SciPy, Scikit-learn, Matplotlib, Seaborn, Plotly, Folium
- Image Processing: ImageJ/Fiji, MATLAB, Gatan Suite
- Environments: JupyterLab/Notebook, Anaconda, Linux, HPC

CERTIFICATIONS & PROFESSIONAL TRAINING

Jan 2026	Supervised Machine Learning: Regression and Classification	*Stanford Online (Coursera)*
Apr 2024	IBM Data Science Professional Certificate	*IBM (Coursera)*
Feb 2024	Python for Everybody Specialization	*University of Michigan (Coursera)*
Jun 2024	Google AI Essentials Certificate	*Google (Coursera)*
May 2023	Teaching Certificate Course for Postgraduate Students (10 weeks)	*Khalifa University — CTL*
Apr 2021	Lab Safety & EHS Certification (comprehensive)	*Khalifa University*
Jun 2021	Hydrofluoric Acid Handling & Chemical Etching SOP	*Khalifa University*
Jul 2020	Responsible Conduct of Research (RCR) — 100% grade	*CITI Program, Florida, USA*

ONLINE PROFILES

Website: shahadathussain.com
LinkedIn: linkedin.com/in/shahadathussain
Google Scholar: scholar.google.com/citations?user=tQNSWaAAAAAJ
ResearchGate: researchgate.net/profile/Shahadat-Hussain
ORCID: orcid.org/0000-0002-4355-2169
GitHub: github.com/drshahadathussain

REFEREES

Available upon request.